SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: B.E / B.Tech

Course Code: MLA02

Course Name: FUNDAMENTALS OF MACHINE LEARNING

Duration: 1:30 Hr

Date : 02-11-2026

Max Marks: 50 Marks

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at $x = 6$. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at $x = 6$.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing ($\alpha = 1$). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

Task 10. AdaBoost starts with five training samples of equal weight. The first weak learner misclassifies samples 2 and 5. Compute the weighted error, the learner weight α, the un-normalised and normalised weights after the update, and analyse how boosting changes the focus of the next learner.

(4) (CO2) (BL4)

Task 11. Two classes share the covariance matrix $\Sigma = [[2, 0.5],[0.5, 1]]$ and have means $\mu_1 = (1, 2)^T$ and $\mu_2 = (3, 1)^T$. (a) Derive the discriminant direction $w = \Sigma^{-1}(\mu_2 - \mu_1)$ and the decision threshold for equal priors. (b) Classify $x = (2.5, 1)$. (c) Evaluate whether the decision changes if the priors are $P(\omega_1) = 0.8$ and $P(\omega_2) = 0.2$.

(4) (CO2) (BL4)

Task 12. A bank evaluates three classifiers on 1,000 test loans (100 actual defaulters). Default is the positive class.

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

A missed defaulter (FN) costs ₹50,000 and a wrongly rejected good applicant (FP) costs ₹5,000.

(a) Compute accuracy, precision, recall and F1-score for each model. [3]

(b) Compute the total expected cost of each model. [2]

(c) Critically evaluate which model should be deployed and explain why accuracy and F1 lead to a different choice than cost. [2]

(d) Propose two modelling or threshold-tuning steps to improve the chosen model and one validation safeguard against over-fitting. [1]

(8) (CO2) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech	Course Code	MLA02
Course Name	Fundamentals of Machine learning.	Date	02-11-2026
Duration	1 Hr 30 Min	Max Marks	50

Register No.: P Surendra Kumar Reddy (192425224)

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	6
Task 4	4	4
Task 5	3	3
Task 6	3	2
Task 7	4	4
Task 8	4	4
Task 9	3	3
Task 10	4	3
Task 11	4	4
Task 12	8	8
TOTAL	**50**	48

Total in words: Fourty eight only

48/50

ASSIGNMENT - II

Name: P. Surendra Kumar Reddy

Reg NO: 19242224

Course Code: MLA0205

Course Name: Fundamentals of Machine Learning.

Task 1. Linear Regression

Given:

X	1	2	3	4	5
Y	2.1	3.9	6.2	7.8	10.1

(a) fit the regression equation using ordinary least squares

The linear regression equation is:

$$y = w_0 + w_1 x$$

Normal equations:

$$\Sigma y = n w_0 + w_1 \Sigma x$$

$$\Sigma xy = w_0 \Sigma x + w_1 \Sigma x^2$$

calculate the required values:

$$\Sigma x = 15, \quad \Sigma y = 30.1$$

$$\Sigma x^2 = 55, \quad \Sigma xy = 110.2$$

sub,

$$30.1 = 5 w_0 + 15 w_1$$

$$110.2 = 15 w_0 + 55 w_2$$

solving the equations:

$$w_1 = \frac{5(110.2) - 15(30.1)}{5(55) - 15^2} = \frac{551 - 451.5}{275 - 225}$$

$$w_1 = \frac{99.5}{50} = \boxed{1.99}$$

Now, $w_0 = \frac{30.1 - 15(1.99)}{5} = \boxed{0.05}$

Therefore, the regression equation is:

$$\boxed{y = 0.05 + 1.99x}$$

(b) predict y at $x = 6$.

$$y = 0.05 + 1.99(6) = 0.05 + 11.94$$

$$\boxed{y = 11.99}$$

(c) Calculates RSS, MSE and R^2,

predicted values:

X	Actual Y	predicted Y	Residual
1	2.1	2.04	0.06
2	3.9	4.03	-0.13
3	6.2	6.02	0.18
4	7.8	8.01	-0.21
5	10.1	10.00	0.10

Residual sum of squares:

$$RSS = \Sigma (y - \hat{y})^2 =$$

$$= 0.0036 + 0.0169 + 0.0324 + 0.0441 + 0.01$$

$$\boxed{RSS = 0.107}$$

mean squared error:

$$MSE = \frac{RSS}{n} = \frac{0.107}{5}$$

$$\boxed{MSE = 0.0214}$$

Coefficient of determination:

$$R^2 = 1 - \frac{RSS}{SST}$$

$$SST = \Sigma (y - \bar{y})^2 = 39.708$$

$$R^2 = 1 - \frac{0.107}{39.708}$$

$$\boxed{R^2 = 0.9973}$$

Analyse the reliability of predicting at $x=6$.

- The R^2 value of 0.9973 indicates an excellent fit.
- The prediction at $x=6$ is reliable because the observed data follows a linear trend.
- However, $x=6$ is outside the training data range.
- Therefore, the prediction is an extrapolation and should be treated with some caution.

Final Answer: Regression equation $= 0.05 + 1.99x$; predicted value $= 11.99$.

Task 2. Ridge Regression

Given: $S_{xx} = 10$, $S_{xy} = 19.9$, $\lambda = 5$

Ridge regression slope formula: $w_1 = \frac{S_{xy}}{S_{xx} + \lambda}$

sub, $w_1 = \frac{19.9}{10 + 5} = \frac{19.9}{15}$

$w_1 = 1.3267$

calculate the percentage shrinkage

ordinary least squares slope:

$$w_{OLS} = \frac{S_{xy}}{S_{xx}} = \frac{19.9}{10} = 1.99$$

percentage shrinkage:

$$= \frac{w_{OLS} - w_{Ridge}}{w_{OLS}} \times 100$$

$$= \frac{1.99 - 1.3267}{1.99} \times 100$$

$$= 33.33\%$$

Bias-variance effect

- Increasing λ reduces the regression coefficients.
- It increases bias and reduces variance.

- It helps prevent overfitting.
- very large λ may cause underfitting.

Final Answer: Ridge slope = 1.3627; percentage shrinkage = 33.3

TASK 3. Naive Bayes classification

Given: 10 labelled emails with two features: offer and link.

- spam emails = 5
- ham emails = 5

(a) calculate priors and conditional probabilities without smoothing

prior probabilities: $P(spam) = \frac{5}{10} = 0.5$

$P(ham) = \frac{5}{10} = 0.5$

Conditional probabilities:

probability	spam	ham
P(offer = Y)	4/5 = 0.8	1/5 = 0.2
P(offer = N)	1/5 = 0.2	4/5 = 0.8
P(Link = Y)	4/5 = 0.8	1/5 = 0.2
P(Link = N)	1/5 = 0.2	4/5 = 0.8

(b) Classify offer = Y, Link = Y.

Using Naive Bayes:

$$P(spam/X) \propto P(spam)\,P(Y/spam)\,P(Y/spam)$$

$$= 0.5 \times 0.8 \times 0.8$$

$$= 0.32$$

for ham:

$$P(ham/X) \propto 0.5 \times 0.2 \times 0.2$$

$$= 0.02$$

Normalizing:

$$P(spam/X) = \frac{0.32}{0.32 + 0.02} = 0.9412$$

$P(spam|X) = 0.9412$

Therefore, the emails is classified as spam.

(c) Repeat using Laplace smoothing ($\alpha = 1$).

Laplace smoothing formula: $P = \frac{Count + \alpha}{Total + \alpha K}$

Here, $K = 2$.

for Spam: $P(offer = Y|spam) = \frac{4+1}{5+2} = \frac{5}{7}$

$P(Link = Y|spam) = 5/7$

for Ham: $P(offer = Y|Ham) = \frac{1+1}{5+2} = \frac{2}{7}$

$P(Link = Y|Ham) = \frac{2}{7}$

posterior probability:

$$P(spam|X) = \frac{0.5(5/7)(5/7)}{0.5(5/7)(5/7) + 0.5(2/7)(2/7)}$$

$$= \frac{25}{25+4} = \frac{25}{29}$$

$P(spam|X) = 0.8621$

Therefore, the emails is classified as spam.

(d) classify offer = N, Link = N with smoothing

for Spam: $P(N|spam) = \frac{2}{7}$

for Ham: $P(N|Ham) = \frac{5}{7}$

$$P(spam|X) = \frac{0.5(2/7)(2/7)}{0.5(2/7)(2/7) + 0.5(5/7)(5/7)} = \frac{4}{4+25}$$

$P(spam|X) = 0.1379$

Therefore, the email is classified as Ham

Analysis:

- The conditional-independence assumption treat of and links as independent given the email class.
- In real emails, these features may be related.
- Laplace smoothing prevents zero probabilities.
- Smoothing reduces the effect of limited training data.

TASK 4. Logistic Regression

Given: $x^{(1)} = (1, 2)$, $y^{(1)} = 1$

$x^{(2)} = (2, 1)$, $y^{(2)} = 0$

$w_1 = 0$, $w_2 = 0$, $b = 0$

Learning rate: $\eta = 0.5$

Step 1: calculate predicted probabilities.

Logistic function: $\sigma(z) = \frac{1}{1+e^{-z}}$

initially, both samples have: $z = 0$

$$p = \frac{1}{1+e^{0}} = 0.5$$

Therefore: $P_1 = P_2 = 0.5$

Step 2: calculate gradients

for mean cross-entropy loss:

$$\frac{\partial L}{\partial w} = \frac{1}{n} \Sigma (p - y) x$$

gradient for w_1:

$$= \frac{1}{2}[(0.5 - 1)(1) + (0.5 - 0)(2)]$$

$$= \frac{1}{2}(-0.5 + 1)$$

$$\frac{\partial L}{\partial w_1} = 0.25$$

gradient for w_2:

$$= \frac{1}{2}[(0.5-1)(2)+(0.5-0)(1)]$$

$$= \frac{1}{2}[-1.0+0.5]$$

$$\boxed{\frac{\partial L}{\partial w_2} = -0.25}$$

gradient for bias:

$$\frac{\partial L}{\partial b} = \frac{1}{2}[-0.5+0.5]$$

$$\boxed{\frac{\partial L}{\partial b} = 0}$$

Step 3: Update parameters

Update formula: $\boxed{w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}}$

$$w_1 = 0 - 0.5(0.25) = -0.125$$

$$w_2 = 0 - 0.5(-0.25) = 0.125$$

$$\boxed{b = 0}$$

Updated parameters:

$$w = (-0.125, 0.125),\ b = 0$$

Step 4: Calculate new predicted probabilities

for sample 1: $z = (-0.125)(1) + (0.125)(2) = 0.125$

$$\boxed{P_1 = \sigma(0.125) = 0.5312}$$

for sample 2: $z = (-0.125)(2) + (0.125)(1) = -0.125$

$$\boxed{P_2 = \sigma(-0.125) = 0.4688}$$

Step 5: Calculate loss before and after updating

Initial loss: $L = -\frac{1}{2}[\ln(0.5) + \ln(1-0.5)]$

$$\boxed{L_{before} = 0.6931}$$

Updated loss: $L = -\frac{1}{2}[\ln(0.5312) + \ln(1-0.4688)]$

$$\boxed{L_{after} = 0.6326}$$

Analysis:

- The predicted probability for the positive sample increased
- The predicted probability for the negative sample decreased
- The loss decreased from 0.6931 to 0.6326.
- Therefore, the gradient-descent update improved the model.

TASK 5. Decision Tree - Entropy and Information Gain.

Given: Total samples = 10

- Yes = 6
- NO = 4

split:

- $A = a_1$: 4 samples (4 yes, 0 NO)
- $A = a_2$: 6 samples (2 yes, 4 NO)

Step 1: Calculate entropy of the parent node.

formula: $$H(S) = -P_1 \log_2 P_1 - P_2 \log_2 P_2$$

$$= -\frac{6}{10} \log_2 \frac{6}{10} - \frac{4}{10} \log_2 \frac{4}{10}$$

$$H(S) = 0.971 \text{ bits.}$$

Step 2: calculate entropy of child nodes.

for a_1: $$H(a_1) = -1 \log_2 1 - 0 \log_2 0$$

$$H(a_1) = 0$$

for a_2: $$H(a_2) = -\frac{2}{6} \log_2 \frac{2}{6} - \frac{4}{6} \log_2 \frac{4}{6}$$

$$H(a_2) = 0.9183$$

Step 3: Calculate weighted entropy

$$H_{split} = \frac{4}{10}(0) + \frac{6}{10}(0.9183)$$

$$H_{split} = 0.551$$

ep 4: calculate Information gain.

$$IG = H(S) - H_{split} = 0.971 - 0.551$$

$$IG = 0.420$$

step 5: calculate a Gini impurity decrease

parent gini: $G(S) = 1 - (0.6)^2 - (0.4)^2$

$$G(S) = 0.48$$

child a_1: $G(a_1) = 1 - 1^2 - 0^2 = 0$

child a_2: $G(a_2) = 1 - (2/6)^2 - (4/6)^2$

$$G(a_2) = 0.4444$$

weighted child Gini:

$$G_{split} = \frac{4}{10}(0) + \frac{6}{10}(0.4444)$$

$$G_{split} = 0.2607$$

Gini decrease:

$$= 0.48 - 0.2667$$

Gini decrease = 0.2133

evaluation:

- Information gain is positive
- Gini impurity decreases after splitting.
- The split separates the samples effectively.
- Therefore, the split is worthwhile.

Task 6. Decision Tree - Cost - Complexity pruning

Given:

- Training samples = 100
- Errors in subtree = 3
- Number of leaves = 4
- Errors after collapsing into one leaf = 8

Cost-complexity formula:

$$R_\alpha(T) = R(T) + \alpha|\text{leaves}|$$

Step 1: calculate subtree cost.

$$R_\alpha(T) = 3 + 4\alpha$$

Step 2: calculate single-leaf cost.

$$R_\alpha(T) = 8 + \alpha$$

Step 3: calculate critical α.

At equal cost:

$$3 + 4\alpha = 8 + \alpha$$
$$4\alpha - \alpha = 8 - 3$$
$$3\alpha = 5$$
$$\alpha = \frac{5}{3} = 1.667$$

Step 5: evaluate for $\alpha = 0.01$ and $\alpha = 0.02$.

for $\alpha = 0.01$

Subtree cost: $3 + 4(0.01) = 3.04$

Single-leaf cost: $8 + 0.01 = 8.01$

since 3.04 is less than 8.01, the subtree should not be pruned.

for $\alpha = 0.02$

Subtree cost: $3 + 4(0.02) = 3.08$

Single-leaf cost: $8 + 0.02 = 8.02$

since 3.08 is less than 8.02, the subtree should ~~not~~ be pruned.

final answer:

- critical $\alpha = 1.667$
- for $\alpha = 0.01$, do not prune.
- for $\alpha = 0.02$, do not prune.

2

Conclusion: The subtree is pre-pruned because it produces fewer training errors and has a lower total cost.

Task 7. Artificial Neural Networks – Back propagation.

Given: $x_1 = 1, x_2 = 2, t = 1$

$w_1 = 0.4, w_2 = 0.3, V = 0.5$

$\eta = 0.5$

Activation function:

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Step 1: forward propagation

hidden layer input: $z_h = w_1 x_1 + w_2 x_2$

$= (0.4)(1) + 0.3(2)$

$= 1$

Hidden layer output: $h = \sigma(1)$

$h = 0.7311$

output layer input: $z_o = Vh$

$= 0.5(0.7311) = 0.3655$

predicted output: $\hat{y} = \sigma(0.3655)$

$\hat{y} = 0.5904.$

step 2: calculate squared-error loss.

$$e = \frac{1}{2}(t-y)^2 = \frac{1}{2}(1-0.5904)^2 = 0.0839$$

$$\boxed{e = 0.0839}$$

step 3: calculate output error term.

using the convention $\boxed{\delta_o = (t-y)\sigma'(z_o)}$

$$\delta_o = (t-y)\,y(1-y)$$

$$= (1-0.5904)(0.5904)(1-0.5904)$$

$$\boxed{\delta_o = 0.09906}$$

step 4: calculate hidden error term

$$\boxed{\delta_h = \delta_o \times v \times h(1-h)}$$

$$= 0.9906 \times (0.5)(0.7311)(1-0.7311)$$

$$\boxed{\delta_h = 0.009738}$$

step 5: calculate gradients

for output weights:

$$\boxed{\frac{\partial e}{\partial v} = -\delta_o h}$$

$$= -0.09906(0.7311)$$

$$\boxed{\frac{\partial e}{\partial v} = -0.07247}$$

for first hidden weight:

$$\boxed{\frac{\partial e}{\partial w_1} = -\delta_h x_1} = (0.009738)(1)$$

$$\boxed{\frac{\partial e}{\partial w_1} = -0.009738}$$

for second hidden weight:

$$\boxed{\frac{\partial e}{\partial w_2} = -\delta_h x_2} = -0.009738 \times 2$$

$$\boxed{\frac{\partial e}{\partial w_2} = -0.019476}$$

Step 6 : Update weights

Update formula : $w_{new} = w_{old} - \frac{\partial \epsilon}{\partial w}$

output weight : $v = 0.5 - 0.5(-0.07242)$

$v = 0.53621$

first hidden weight : $w_1 = 0.4 - 0.5(-0.009738)$

$w_1 = 0.40487$

second hidden weight : $w_2 = 0.3 - 0.5(-0.019476)$

$w_2 = 0.30974$

Step 7 : Calculate new loss

new hidden output : $h' = \sigma(0.40487 + 0.30974(2))$

$h' = 0.73582$

New predicted output : $y' = \sigma(0.53621 \times 0.73582)$

$y' = 0.59738$

New loss : $\epsilon' = \frac{1}{2}(1 - 0.59738)^2$

$\epsilon' = 0.08105$

final Answer :

parameter	updated value
w_1	0.40487
w_2	0.30974
v	0.53621
New loss	0.08105

Conclusion : Backpropagation updates the network weights and reduces the error from 0.0839 to 0.08105.

Task 8. Support vector machine (SVM)

Given: $w = (0.5, 0.5)$, $b = -2$

Support vectors: $(3,3) \rightarrow 1$

$(1,1) \rightarrow -1$

Decision function: $f(x) = w^T x + b$

(a) Verify support vectors and calculate margin width.

for positive support vector $(3,3)$:

$$f(x) = 0.5(3) + 0.5(3) - 2$$

$$= 1$$

Therefore, it lies on the positive margin hyperplane.

for negative support vector $(1,1)$:

$$f(x) = 0.5(1) + 0.5(1) - 2$$

$$= -1$$

Therefore, it lies on the negative margin hyperplane.

calculate weight norm:

$$\|w\| = \sqrt{0.5^2 + 0.5^2} = \sqrt{0.5} = 0.7071$$

margin width:

$$\frac{2}{\|w\|} = \frac{2}{0.7071} =$$

margin width = 2.8284

(b) classify the points and calculate distances

point $(4,2)$:

$$f(x) = 0.5(4) + 0.5(2) - 2$$

$$= 2 + 1 - 2 = 1$$

since the result is positive, it belongs to class -1

Distance:

$$d = \frac{|f(x)|}{\|w\|} = \frac{1}{0.7071} = 1.4142$$

$d = 1.4142$

oint (1,2): $f(x) = 0.5(1) + 0.5(2) - 2$

$= 0.5 + 1 - 2 = \boxed{-0.5}$

since the result is negative, it belong to class -1

Distance: $d = \frac{|-0.05|}{0.7071} \Rightarrow \boxed{d = 0.7071}$

(c) calculate hinge loss

formula: $L = \max(0, 1 - yf(x))$

for (1,2), true label = -1:

$L = \max(0, 1 - (-1)(-0.5))$

$= \max(0, 0.5)$

$\boxed{L = 0.5}$

for (2,2), true label = +1:

$f(x) = 0.5(2) + 0.5(2) - 2 = 0$

$L = \max(0, 1 - (+1)(0))$

$\boxed{L = 1}$

conclusion:

- Both points violate the margin.
- point (1,2) is correctly classified but lies inside the margin.
- point (2,2) lies directly on the decision boundary.

Task 9. Ensemble Learning - Majority voting

Given:

Accuracy of each independent classifier: $p = 0.70$

error probability: $q = 1 - p = 0 - 0.70 = 0.30$

Step 1: calculate ensemble accuracy for 3 classifiers

majority voting formula: $\boxed{p = \binom{3}{3} p^3 + \binom{3}{2} p^2 q}$

$= (0.7)^3 + 3(0.7)^2(0.3)$

$= 0.343 + 0.441$

$P = 0.784 = 78.4\ \%$

Step 2: calculate ensemble accuracy for 5 classifie

$$P = \binom{5}{5} p^5 + \binom{5}{4} p^4 q + \binom{5}{3} p^3 q^2$$

$$= (0.7)^5 + 5(0.7)^4(0.3) + 10(0.7)^3(0.2)^2$$

$$= 0.16807 + 0.36015 + 0.3087$$

$$P = 0.83692 = 83.692\ \%$$

Step 3: Analysis the result

classifiers	ensemble accuracy
1	70 %
3	78.4 %
5	83.692 %

- Majority voting improves classification accuracy.
- Increasing the number of independent classifiers can improve performance.
- strongly correlated errors reduce the benefits of ensemble learning.
- If classifiers make similar mistakes, majority voting may not improve accuracy significantly.

conclusion: five independent classifiers achieve better accuracy than three classifiers.

Task 10 – Ada Boost Algorithm

Given:

- Total training samples = 5
- Initial weight of each sample = $\frac{1}{5} = 0.2$
- missclassified samples = 2 & 5

Step 1: calculate weighted error

formula:

$$\epsilon = \sum_{i; y_i \neq h(w_i)} w_i \Rightarrow 0.2 + 0.2$$

$$\epsilon = 0.4$$

Step 2: calculate learner weights α

formula:

$$\alpha = \frac{1}{2} \ln\left(\frac{1-\epsilon}{\epsilon}\right) = \frac{1}{2} \ln\left(\frac{1-0.4}{0.4}\right)$$

$$= \frac{1}{2} \ln(1.5)$$

$$\alpha = 0.2027$$

step 3: calculate updated un-normalised weights

for correctly classified samples:

$$w_{correct} = 0.2 e^{-\alpha} = 0.2 e^{-0.2027}$$

$$w_{correct} = 0.1633$$

for missclassified samples:

$$w_{wrong} = 0.2 e^{\alpha} = 0.2 e^{0.2027}$$

$$w_{wrong} = 0.2449$$

Un-normalised weights.

sample	classification	weights.
1	correct	0.1633
2	wrong	0.2449
3	correct	0.1633
4	correct	0.1633
5	wrong	0.2449

Step 4: Calculate normalised weights.

Sum of weights: $Z = 3(0.1633) + 2(0.2449)$

$$Z = 0.9798$$

Normalised weights: $\boxed{w_i' = \frac{w_i}{Z}}$

for correctly classified samples

$$w' = \frac{0.1633}{0.9798}$$

$$w' = 0.1667$$

for misclassified samples

$$w' = \frac{0.2449}{0.9798}$$

$$w' = 0.25$$

3 final weights.

Sample	Normalised weights.
1	0.1667
2	0.2500
3	0.1667
4	0.1667
5	0.2500
Total	1.0000

Step 5: Analyse the effects of boosting

- AdaBoost increases the weights of misclassified samples.
- It decreases the weights of correctly classified samples.
- The next weak learner focuses more on previously misclassified samples.
- Repeated boosting improves the overall classification performance.

Conclusion: AdaBoost gives more importance to difficult samples so that next learner can improve classification.

Q11 Linear Discriminant Analysis (LDA)

Given: Covariance matrix:

$$\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

mean vectors:

$$\mu_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mu_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

(a) Derive discriminant direction and decision method.

step 1: calculate mean difference

$$\mu_2 - \mu_1 = \begin{bmatrix} 3-1 \\ 1-2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

step 2: calculate inverse covariance matrix

$$\Sigma^{-1} = \frac{1}{2(1)-(0.5)(0.5)} \begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}$$

$$= \frac{1}{1.75} \begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}$$

$$\Sigma^{-1} = \begin{bmatrix} 0.5714 & -0.2857 \\ -0.2857 & 1.1429 \end{bmatrix}$$

step 3: calculate discriminant direction

formula: $\omega = \Sigma^{-1}(\mu_2 - \mu_1)$

$$= \begin{bmatrix} 0.5714 & -0.2857 \\ -0.2857 & 1.1429 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\omega = \begin{bmatrix} 1.4286 \\ -1.7143 \end{bmatrix}$$

step 4: calculate decision threshold for equal priors.

$$\omega^T x_0 = \omega^T \frac{\mu_1 + \mu_2}{2}$$

$$\frac{\mu_1 + \mu_2}{2} = \begin{bmatrix} 2 \\ 1.5 \end{bmatrix}$$

$$\omega^T x_0 = (1.4286)(2) + (-1.7143)(1.5)$$

$$= 2.8572 - 2.57145$$

Threshold = 0.2857

(b) classify $X = (2.5, 1)$

calaulate discriminant score: $g(x) = w^T x$

$= (1.4286)(2.5) + (-1.7143)(1)$

$= 3.5715 - 1.7143$

$g(x) = 1.8572$

since: $1.8572 > 0.2857$

The sample belongs to class ω_2

(c) evaluate the effect of unequal priors.

Given: $P(\omega_1) = 0.8$ $P(\omega_2) = 0.2$

new decision threshold: $\text{Threshold} = 0.2857 + \ln\left(\frac{0.8}{0.2}\right)$

$= 0.2857 + \ln(4)$

$\text{Threshold} = 1.6720$

Compare: $1.8572 > 1.6720$

Therefore, the sample still belong to class ω_2.

Conclusion:

- The discriminant direction is $(1.4286, -1.7143)$
- Equal-prior threshold $= 0.2857$
- Unequal-prior threshold $= 1.6720$
- The classification does not change, even though the decision threshold increases.

Task 12. Classifier evaluation and Cost-Based model selection

Given:

Total test loans = 1000

Actual defaulters = 100

Cost:

- False Negative (FN) = ₦ 50,000
- False positive (FP) = ₦ 5,000

Confusion matrix:

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

(a) calculate accuracy, precision, recall and F1-score.

formulas:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2TP}{2TP + fp + FN}$$

Model A:

Accuracy : $\frac{70 * 810}{70 + 30 + 90 + 810} = \frac{880}{1000} \times 100 =$ **88%**

precision : $\frac{70}{70 + 90} \times 100 =$ **43.75%**

Recall : $\frac{70}{70 + 30} \times 100 =$ **70%**

F1-score : $\frac{2(70)}{140 + 90 + 30} \times 100 =$ **53.85%**

Model B:

Accuracy : $\frac{85+720}{1000} \times 100 = 80.5\%$

precision : $\frac{85}{85+180} \times 100 = 32.08\%$

Recall : $\frac{85}{85+15} \times 100 = 85\%$

F1-Score : $\frac{2(85)}{170+180+15} \times 100 = 46.58\%$

Model C:

Accuracy : $\frac{60+870}{1000} \times 100 = 93\%$

precision : $\frac{60}{60+30} \times 100 = 66.67\%$

Recall : $\frac{60}{60+40} \times 100 = 60\%$

F1-Score : $\frac{2(60)}{120+30+40} \times 100 = 63.16\%$

Comparision table :

Metric	model A	model B	model C
Accuracy	88%	80.5%	93%
precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%
F1-Score	53.85%	46.58%	63.16%

(b) Calculate total expected cost of each model.

formula : Total Cost = (FN X 50000) + (FP * 5000)

Model A

= (30 X 50000) + (90 X 5000)

= 1500000 + 450000

$Cost_A$ = ₹ 19,50,000

Model B :

$= (15 \times 50000) + (180 \times 5000)$

$= 750000 + 900000$

$\text{Cost}_B = ₦16,50,000$

model C :

$(40 \times 50000) + (30 \times 5000)$

$= 2000000 + 150000$

$\text{Cost}_C = ₦21,50,000$

Comparision:

Model	Total expected cost
A	₦19,50,000
B	₦16,50,000
C	₦21,50,000

(C) Critically evaluate which model should be deployed

Model B should be deployed based on the given cost structure.

Reasons:

- Model B has the highest recall of 85%.
- It identifies the highest number of actual defaulters.
- It has the lowest total expected cost of ₦16,50,000.
- Model C has the highest accuracy (93%) and F1-Score (63.16%) but it misses 40 actual defaulters.
- Missing a defaulter cost is ₦50,000, which is much higher than wrongly rejecting a good applicant (₦5,000).

Therefore, Cost-based evaluation selects Model B, whereas accuracy and f1-score select model C.

(d) Suggest two improvements and one validation safeguard.

Two modelling or threshold-tuning steps.

1. threshold tuning: adjust the classification threshold to balance false negatives and false positives while reducing total cost.

2. cost-sensitive learning: Give higher importance to defaulters during training so that the model identifies more risky loans.

validation safeguard:

- use stratified cross-validation and independent test dataset to check model performance and prevent overfitting.

Final conclusion: Model B is the best choice because it gives the lowest expected financial cost, even though its accuracy and f1-score are lower than model-C.